

VULNERABILITY ASSESSMENT AND PENETRATION TESTING (VAPT)

Using PortIntel in an Isolated Metasploitable Security Lab

Authorized Target: Metasploitable 2 — 192.168.189.130

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Field	Value
Project Type	Vulnerability Assessment and Penetration Testing (VAPT)
Environment	Authorized Local Cybersecurity Laboratory
Primary Custom Tool	PortIntel v2.0
Independent Validation Tool	Nmap
Testing Platform	Kali Linux (Assessment Workstation)
Target Platform	Metasploitable 2 (Intentionally Vulnerable)
Target IP Address	192.168.189.130
Virtualization	VMware (Host-Only Network)
Web Application Observed	OWASP Mutillidae
Assessment Status	Reconnaissance, Vulnerability Identification, & Controlled Validation Completed

1. Executive Summary

This report documents an authorized Vulnerability Assessment and Penetration Testing (VAPT) engagement performed entirely within an isolated virtual laboratory. The laboratory consists of a Kali Linux assessment workstation and an intentionally vulnerable Metasploitable 2 target, connected through a VMware Host-Only virtual network with no external exposure.

The assessment used **PortIntel v2.0**, a custom Python-based network reconnaissance and vulnerability-intelligence framework, as the primary tool for host discovery, port and service enumeration, service fingerprinting, and vulnerability intelligence. Nmap was subsequently used as an independent validation tool to cross-check PortIntel's findings and to perform vulnerability-specific technical validation.

PortIntel identified 12 open TCP ports on the target and fingerprinted the associated services. Independent Nmap verification confirmed the service inventory, and a vulnerability-oriented Nmap assessment identified a range of candidate vulnerability intelligence associated with the discovered software versions.

The assessment produced one confirmed, technically validated critical finding: a remotely exploitable backdoor in **vsFTPD 2.3.4 (CVE-2011-2523)**, which was demonstrated to allow root-level command execution on the target within the authorized laboratory. A manual authentication-bypass test against a Mutillidae login endpoint, performed to validate an automated SQL-injection indication, did not reproduce the behavior and is documented as a negative

result rather than a confirmed vulnerability.

This report distinguishes throughout between PortIntel discovery and fingerprinting, independent Nmap verification, scanner-reported candidate vulnerability intelligence, and technically validated vulnerabilities, in line with a professional, evidence-based VAPT methodology.

2. Project Objectives

- Build a safe and isolated penetration-testing laboratory.
- Identify reachable systems within the authorized network.
- Discover exposed network services on the authorized target.
- Identify software and service versions where reliably available.
- Perform banner-based service fingerprinting.
- Identify potential vulnerabilities associated with discovered services.
- Correlate software information with CVE intelligence.
- Prioritize identified vulnerabilities using standardized risk information.
- Independently validate important findings using Nmap.
- Demonstrate selected findings safely within the authorized laboratory.
- Document technical evidence and remediation recommendations.
- Demonstrate the practical application of PortIntel within a professional VAPT workflow.

3. Ethical and Authorization Statement

All security testing documented in this report was performed exclusively against systems owned and controlled within an isolated virtual laboratory.

Metasploitable 2 is intentionally vulnerable software deployed specifically for security training and penetration-testing purposes.

No public systems, third-party infrastructure, university infrastructure, home networks, or production systems were included within the assessment scope.

The assessment did not include denial-of-service testing, destructive payloads, uncontrolled malware deployment, or unauthorized persistence.

4. Scope and Rules of Engagement

4.1 In-Scope Systems

Assessment Workstation: Kali Linux virtual machine running PortIntel and supporting validation tools.

Target System: Metasploitable 2 virtual machine configured as the intentionally vulnerable target (192.168.189.130).

Network: VMware Host-Only virtual network connecting the two systems.

4.2 Authorized Activities

- Host discovery and network reconnaissance
- TCP/UDP port scanning and service enumeration
- Banner grabbing and service fingerprinting

- Vulnerability identification and CVE intelligence analysis
- Risk prioritization and independent vulnerability verification
- Controlled non-destructive validation and evidence collection
- Security reporting

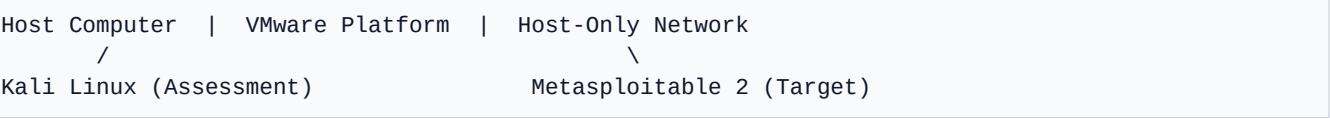
4.3 Out-of-Scope Activities

- Public Internet targets or third-party systems
- University/college infrastructure or home network devices
- Denial-of-service attacks or destructive payload execution
- Uncontrolled malware deployment or unauthorized persistence
- Testing outside the isolated laboratory

5. Laboratory Architecture

The assessment laboratory uses an isolated architecture: **Kali Linux (Security Assessment Workstation) → VMware Host-Only Network → Metasploitable 2 (Authorized Vulnerable Target)**.

The Host-Only configuration allows Kali Linux and Metasploitable 2 to communicate while preventing the intentionally vulnerable target from being directly exposed to an external network.



6. Tools and Technologies

6.1 PortIntel v2.0

PortIntel is the primary custom security tool used during this project. It is a Python-based network reconnaissance and vulnerability-intelligence framework designed to assist security analysts with host discovery, TCP/UDP port scanning, service identification, banner grabbing, version fingerprinting, CPE resolution, NVD CVE intelligence, CVSS-based risk prioritization, MITRE ATT&CK; contextualization, and structured assessment reporting.

PortIntel separates vulnerability severity from general service-exposure concerns, helping analysts distinguish known software vulnerabilities from potentially unsafe network configurations. PortIntel is an independently developed cybersecurity engineering project; it was applied here as the reconnaissance and vulnerability-intelligence component of a broader, tool-assisted VAPT workflow, alongside industry-standard tools used for independent validation.

6.2 Supporting Technologies

Kali Linux: Used as the security assessment workstation. It provides the environment from which PortIntel and supporting validation tools are executed.

Metasploitable 2: Used as the intentionally vulnerable target machine. It provides multiple deliberately insecure network services and applications suitable for security-assessment training in an isolated environment.

VMware: Used to virtualize both the assessment workstation and the vulnerable target. A Host-Only network configuration isolates the vulnerable environment from external networks.

7. Assessment Methodology

The project follows a 13-phase assessment workflow: **Authorization & Scope** → **Laboratory Isolation** → **Connectivity Verification** → **PortIntel Host Discovery** → **PortIntel Port Enumeration** → **Service Fingerprinting** → **Vulnerability Intelligence** → **Nmap Independent Verification** → **Vulnerability-Specific Assessment** → **Controlled Technical Validation** → **Evidence Collection** → **Risk Classification** → **Remediation Recommendations & Reporting**.

At the time of this report, environment preparation, connectivity verification, PortIntel deployment, host discovery, port and service enumeration, service fingerprinting, independent Nmap verification, vulnerability identification, and controlled validation of the critical finding have all been completed. Actual remediation implementation has not been performed and remains a recommended next step.

8. Environment Setup & Connectivity

Metasploitable 2 was imported into VMware and configured as the authorized vulnerable target, with its network adapter attached to the isolated VMware Host-Only network. After starting the system, the network configuration was inspected from the Metasploitable terminal (ifconfig), confirming assigned IP address **192.168.189.130**. Kali Linux was similarly configured on the same Host-Only subnet (ip addr). Basic connectivity was verified via `ping -c 4 192.168.189.130`.

9. PortIntel Deployment & Verification

PortIntel v2.0 was deployed on the Kali Linux assessment workstation inside an isolated Python virtual environment using an editable install sequence:

```
git clone https://github.com/kharbashpriyanshu/Port_Intel.git
cd Port_Intel
python3 -m venv .venv && source .venv/bin/activate
python3 -m pip install -e .
portintel version
```

Expected output confirmed: PortIntel Version 2.0.0, Python 3.13.11, running on Linux 6.18.5+kali-amd64 (64-bit).

10. Host Discovery

PortIntel's discovery functionality was used to identify reachable systems within the authorized Host-Only laboratory network (`portintel discover --network 192.168.189.0/24`). PortIntel host discovery completed successfully, reporting four alive hosts on the laboratory subnet, including the gateway (192.168.189.2), two Kali Linux interface addresses, and the Metasploitable 2 target (192.168.189.130).

11. Port & Service Enumeration

Port and service enumeration is an important stage of VAPT because every exposed service increases the potential attack surface of a system. PortIntel v2.0 was used as the primary reconnaissance tool during this phase (`portintel scan --target 192.168.189.130`).

PortIntel evaluated 1,024 TCP ports on the target and identified **12 open ports** in a scan completing in **2.81 seconds**. The discovered open ports represent externally reachable services within the laboratory network and form the primary attack surface considered during subsequent vulnerability assessment activities.

Port	Service	State	Initial Security Observation
21	FTP	Open	FTP service detected; version information exposed via banner fingerprinting.
22	SSH	Open	Remote administration service identified with OpenSSH fingerprint information.
23	Telnet	Open	Clear-text remote-access protocol identified and flagged as an exposure concern.
25	SMTP	Open	Postfix SMTP service detected.
53	DNS	Open	DNS service accessible from the assessment network.
80	HTTP	Open	Web service detected; HTTP banner information obtained.
111	SUNRPC	Open	RPC service exposed to the assessment network.
139	NetBIOS-SSN	Open	NetBIOS/SMB-related service exposed.
445	Microsoft-DS	Open	SMB-related Microsoft-DS service exposed.
512	EXEC	Open	Legacy remote execution service exposed.
513	LOGIN	Open	Legacy remote login service exposed.
514	SHELL	Open	Legacy remote shell service exposed.

```

(.venv)-(Eagle@Eagle)-[~/Port_Intel/Port_Intel]
$ portintel scan --target 192.168.189.130
=====
PORTINTEL v2.0 - Intelligent Network Reconnaissance
=====
Scanning 192.168.189.130...

=====
SCAN SUMMARY FOR 192.168.189.130
=====
Total Time : 2.81 seconds
Scanned : 1024 ports
Open Ports : 12

-----
PORT  SERVICE      RISK    STATUS
-----
21    FTP           Medium  OPEN
|- Version: 2.3.4
|- CPE: cpe:2.3:a:beasts:vsftpd:2.3.4:*:*:*:*:*
|- MITRE: T1048 - Exfiltration Over Alternative Protocol
|- Banner: 220 (vsFTPd 2.3.4)
22    SSH           Info    OPEN
|- Version: 2.0-OpenSSH
|- CPE: cpe:2.3:a:openssh:openssh:2.0-openssh:*:*:*:*:*
|- MITRE: T1021 - Remote Services (Lateral Movement)
|- Banner: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
23    TELNET        Medium  OPEN
|- MITRE: T1021 - Unencrypted Remote Services
|- Banner: #
25    SMTP           Info    OPEN
|- CPE: cpe:2.3:a:postfix:postfix:*:*:*:*:*
|- Banner: 220 metasploitable.localdomain ESMTD Postfix (Ubuntu)
53    DOMAIN         Info    OPEN
80    HTTP           Info    OPEN
|- Version: 1.1
|- CPE: cpe:2.3:a:apache:http_server:1.1:*:*:*:*
|- Banner: HTTP/1.1 200 OK Date: Sun, 26 Jul 2026 18:07:38 GMT Serve...

111   SUNRPC         Info    OPEN
139   NETBIOS-SSN   Info    OPEN
445   MICROSOFT-DS   Info    OPEN
512   EXEC           Info    OPEN
|- Banner: Where are you?
513   LOGIN         Info    OPEN
514   SHELL         Info    OPEN
=====
Total Time : 2.81 seconds
Scanned : 1024 ports
Open Ports : 12

-----
PORT  SERVICE      RISK    STATUS
-----
21    FTP           Medium  OPEN
|- Version: 2.3.4
|- CPE: cpe:2.3:a:beasts:vsftpd:2.3.4:*:*:*:*:*
|- MITRE: T1048 - Exfiltration Over Alternative Protocol
|- Banner: 220 (vsFTPd 2.3.4)
22    SSH           Info    OPEN
|- Version: 2.0-OpenSSH
|- CPE: cpe:2.3:a:openssh:openssh:2.0-openssh:*:*:*:*:*
|- MITRE: T1021 - Remote Services (Lateral Movement)
|- Banner: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
23    TELNET        Medium  OPEN
|- MITRE: T1021 - Unencrypted Remote Services
|- Banner: #
25    SMTP           Info    OPEN
|- CPE: cpe:2.3:a:postfix:postfix:*:*:*:*:*
|- Banner: 220 metasploitable.localdomain ESMTD Postfix (Ubuntu)
53    DOMAIN         Info    OPEN
80    HTTP           Info    OPEN
|- Version: 1.1
|- CPE: cpe:2.3:a:apache:http_server:1.1:*:*:*:*
|- Banner: HTTP/1.1 200 OK Date: Sun, 26 Jul 2026 18:07:38 GMT Serve...

111   SUNRPC         Info    OPEN
139   NETBIOS-SSN   Info    OPEN
445   MICROSOFT-DS   Info    OPEN
512   EXEC           Info    OPEN
|- Banner: Where are you?
513   LOGIN         Info    OPEN
514   SHELL         Info    OPEN
=====

```

Figure 10: PortIntel v2.0 identifying the exposed TCP attack surface of the authorized Metasploitable 2 target.

12. Service Fingerprinting

Where sufficient information was available, PortIntel used service banners and version information to improve fingerprint accuracy and, where supported, resolved CPE identifiers. Values that PortIntel could not reliably determine are recorded as Unknown/Unresolved rather than guessed.

Port	Service	Product	Version	Banner / Fingerprint
21	FTP	vsFTPD	2.3.4	220 (vsFTPD 2.3.4)
22	SSH	OpenSSH	4.7p1	SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
23	Telnet	Unknown/Unresolved	Unknown/Unresolved	#
25	SMTP	Postfix	Unknown/Unresolved	220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
80	HTTP	Apache httpd	Unknown/Unresolved	HTTP/1.1 200 OK

PortIntel additionally annotated each service with MITRE ATT&CK; context (for example, flagging Telnet as an unencrypted remote-services exposure, and FTP/SSH as potential lateral-movement or remote-services vectors) and a qualitative risk rating.

13. Independent Nmap Verification

Following PortIntel's port and service enumeration, an independent verification step was performed using Nmap (nmap -sV -p 21,22,23,25,53,80,111,139,445,512,513,514 192.168.189.130). The purpose of this stage was not to replace PortIntel, but to cross-validate the network exposure and service-fingerprinting results produced by the custom reconnaissance framework.

Nmap independently confirmed all 12 open ports and their associated services, and additionally reported product/version detail for services where PortIntel had recorded Unknown/Unresolved (for example, identifying Telnet as Linux telnetd and NetBIOS/Microsoft-DS as Samba smbd 3.X–4.X). Using PortIntel as the primary reconnaissance framework with Nmap as an independent verification mechanism provides stronger technical evidence than relying on a single scanner.

14. Vulnerability Identification

Following independent verification of the exposed services, the assessment progressed from network reconnaissance to vulnerability identification using the evidence chain: **Open Port** → **Service** → **Product/Version** → **Vulnerability Intelligence** → **CVE/CVSS** → **Technical Validation**.

An additional vulnerability-oriented Nmap assessment was performed against selected services on the authorized Metasploitable 2 target (nmap -sV --script vuln -p 21,22,23,25,53,80,111,139,445 192.168.189.130).

IMPORTANT NOTE ON THE VULNERS OUTPUT: The large number of high-CVSS CVE/exploit references returned by the Nmap Vulners script (for OpenSSH 4.7p1 in particular) represent version-to-database correlations, not individually validated vulnerabilities. Several of the referenced CVE identifiers correspond to software released after this specific OpenSSH build and are included by the Vulners engine on the basis of broad version-string matching. These entries are recorded here as candidate vulnerability intelligence only, and are not claimed as confirmed findings.

15. Confirmed Finding — VAPT-001

15.1 VAPT-001 — vsFTPD 2.3.4 Backdoor

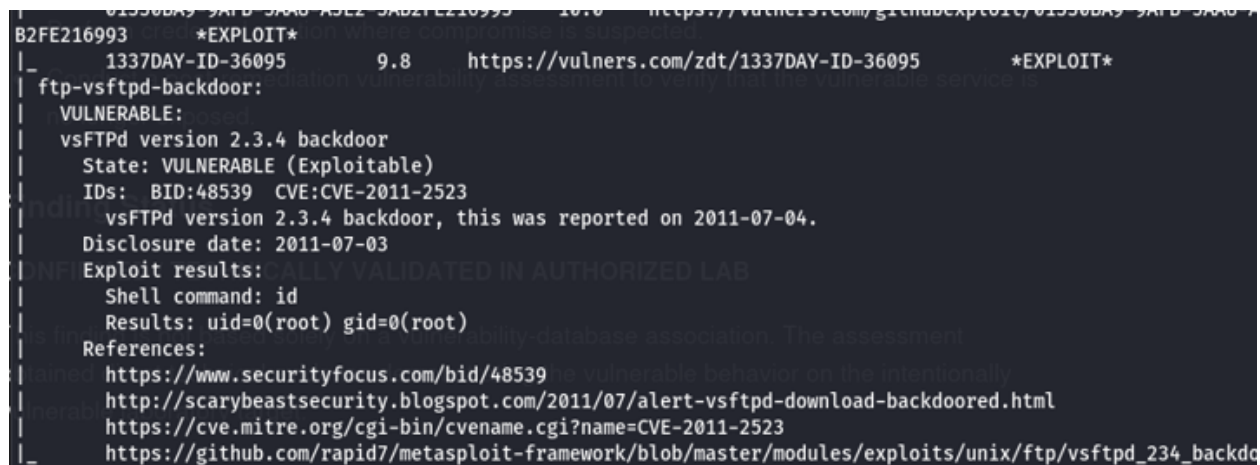
Attribute	Assessment Result
Finding ID	VAPT-001
Target	192.168.189.130 (Metasploitable 2)
Port / Service	21/TCP — FTP (vsFTPD 2.3.4)
CVE / Risk	CVE-2011-2523 — Critical (CVSS v3: 10.0)
Detection Source	PortIntel (discovery/fingerprinting) + Nmap NSE (ftp-vsftpd-backdoor)
Validation Result	Nmap NSE ftp-vsftpd-backdoor — State: VULNERABLE (Exploitable)
Demonstrated Access	uid=0(root) gid=0(root) — Root-Level System Compromise
Status	Confirmed / Technically Validated in Authorized Lab

Technical Description & Evidence:

PortIntel identified TCP/21 as open and fingerprinted the FTP service as vsFTPD 2.3.4 during port and service enumeration. Nmap independently confirmed this service and version. During vulnerability-specific validation, the Nmap Scripting Engine identified the service as affected by the known vsFTPD 2.3.4 backdoor vulnerability associated with CVE-2011-2523.

Unlike a version-only vulnerability match, the NSE validation reported the target as State: VULNERABLE (Exploitable) and executed the non-destructive identification command `id`, which returned `uid=0(root) gid=0(root)`.

To be clear: PortIntel identified and fingerprinted the exposed FTP service; it did not itself exploit the vulnerability. Nmap independently confirmed the service and version, and Nmap's vulnerability-specific NSE validation demonstrated that the vsFTPD 2.3.4 backdoor was exploitable.



```

B2FE216993 *EXPLOIT*
|_ 1337DAY-ID-36095 9.8 https://vulners.com/zdt/1337DAY-ID-36095 *EXPLOIT*
| ftp-vsftpd-backdoor:
| VULNERABLE:
| vsFTPD version 2.3.4 backdoor
| State: VULNERABLE (Exploitable)
| IDs: BID:48539 CVE:CVE-2011-2523
| vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
| Disclosure date: 2011-07-03
| Exploit results:
| Shell command: id
| Results: uid=0(root) gid=0(root)
| References:
| https://www.securityfocus.com/bid/48539
| http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
| https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdo

```

Figure 13: Confirmed ftp-vsftpd-backdoor NSE finding, showing CVE-2011-2523, VULNERABLE (Exploitable) state, and root shell output.

Security Impact:

Remote command execution and root-level system compromise. An attacker obtaining this level of access could

access protected information, modify system configuration, establish persistence, or use the compromised host as a pivot point.

Recommended Remediation:

Remove the vulnerable vsFTPD 2.3.4 installation immediately. Install a currently supported FTP/SFTP solution from a trusted, authenticated repository, or disable FTP entirely if not operationally required. Prefer encrypted file-transfer mechanisms such as SFTP.

16. Additional Security Observations and Candidate Findings

The vulnerability-oriented scan reported a number of additional security observations, summarized below according to the strength of supporting evidence. None of the items in this section are presented as confirmed exploits.

Observation	Classification
SSL/TLS-related weaknesses (e.g., POODLE/CVE-2014-3566-class)	Candidate vulnerability — requires further validation
Weak Diffie-Hellman configuration / Logjam-related observations	Candidate vulnerability — requires further validation
ISC BIND vulnerability intelligence (DNS service)	Candidate vulnerability — requires further validation
Multiple outdated service/version associations (OpenSSH 4.7p1)	Candidate vulnerability — version-to-database correlation only
Clear-text Telnet exposure (TCP/23)	Confirmed configuration weakness / exposure concern
Legacy EXEC/LOGIN/SHELL (r-services on 512/513/514)	Confirmed configuration weakness / exposure concern
SMB/NetBIOS exposure (ports 139/445)	Exposure concern — requires further validation
Web application attack surface (OWASP Mutillidae on TCP/80)	Exposure concern
Exposed /phpinfo.php	Confirmed exposure concern — information disclosure
Exposed /phpMyAdmin/	Confirmed exposure concern — administrative interface exposure
Directory listing / information exposure	Exposure concern — requires further validation
Potential SQL injection locations reported within Mutillidae	Candidate vulnerability — see Section 17 for manual validation result

17. Manual SQL Injection Validation / False-Positive Handling

17.1 Finding VAPT-002 — Reported SQL Injection in Mutillidae

Attribute	Assessment Result
Finding ID	VAPT-002
Target / Service	192.168.189.130 — HTTP (TCP/80) OWASP Mutillidae
Vulnerability Type	SQL Injection (automated indication)
Detection Source	Automated vulnerability-oriented scan + manual validation
Status	Not Confirmed — Negative Manual Validation Result on Tested Endpoint

During the vulnerability-oriented reconnaissance phase, automated assessment reported possible SQL injection locations within the Mutillidae web application. Rather than treating this automated result as sufficient proof, manual validation was performed against a specific login endpoint to determine whether crafted input could actually alter application behavior.

A normal application login request was first submitted to establish expected baseline behavior (Username: test | Password: test). Following the baseline attempt, a controlled SQL injection authentication-bypass payload was submitted (Username: ' OR '1'='1' -- | Password: test).

Result: The manipulated authentication request produced **no observable authentication bypass**. The application remained on the login interface, consistent with the baseline request. Therefore, this specific login test did not provide sufficient evidence to classify the tested login endpoint as manually confirmed SQL injectable.

This negative validation result is retained in this report because it demonstrates professional false-positive handling and evidence-based vulnerability classification: automated scanner output was manually tested rather than automatically accepted as a confirmed finding.

18. Role of PortIntel in the VAPT Assessment

PortIntel served as the project's primary custom reconnaissance and vulnerability-intelligence framework. Its demonstrated contribution included authorized host discovery, TCP port enumeration, service identification, banner/service fingerprinting, product/version identification, CPE resolution, vulnerability intelligence, CVSS risk prioritization, MITRE ATT&CK; contextualization, and structured reporting workflow.

19. Findings Summary

Finding ID	Description	Port	CVE	Severity	Status
VAPT-001	vsFTPD 2.3.4 Backdoor	21/TCP	CVE-2011-2523	Critical	Confirmed / Exploitable in Lab
VAPT-002	Reported SQL Injection in Mutillidae	80/TCP	N/A	N/A	Not Confirmed (negative manual test)
OBS-01	Clear-text Telnet exposure	23/TCP	N/A	Exposure	Confirmed configuration weakness
OBS-02	Legacy r-services (EXEC/LOGIN/SHELL)	512-514	N/A	Exposure	Confirmed configuration weakness
OBS-03	SMB/NetBIOS exposure	139/445	N/A	Exposure	Candidate — requires further validation
OBS-04	Exposed /phpinfo.php & /phpMyAdmin/	80/TCP	N/A	Exposure	Confirmed exposure concern
OBS-05	OpenSSH 4.7p1 version-correlated CVEs	22/TCP	Multiple	Candidate	Candidate — not individually validated

20. Risk Analysis & 21. Remediation Recommendations

Vulnerability severity in this report is based on technical impact and standardized vulnerability information (CVE/CVSS) where such information has been directly validated against the target. This is deliberately distinguished from general service exposure concern, which reflects an inherently risky configuration (such as an unencrypted protocol) without necessarily corresponding to a specific validated CVE.

Recommended Action Plan:

1. Immediately remove the vulnerable vsFTPD 2.3.4 installation (VAPT-001) and replace with a supported SFTP solution.
2. Disable unnecessary legacy remote-access protocols, particularly Telnet (TCP/23) and r-services (TCP/512-514).
3. Patch and update exposed network services (OpenSSH, Apache, Samba) to currently supported versions.
4. Harden the web application layer by removing or restricting administrative interfaces (/phpMyAdmin/) and debug endpoints (/phpinfo.php).
5. Perform a full vulnerability rescan after remediation is implemented and independently verify closure.

24. Skills Demonstrated & 25. Limitations

Skills Demonstrated: VAPT methodology, structured assessment workflow, host discovery, TCP enumeration, banner fingerprinting, CVE/CPE correlation, CVSS risk prioritization, independent validation, controlled non-destructive validation, evidence handling, false-positive management, remediation planning, and custom security engineering (PortIntel v2.0).

Limitations: Testing was conducted within an isolated laboratory against an intentionally vulnerable target; only one login endpoint within Mutillidae was manually tested for SQL injection; remediation has not yet been implemented or retested; several candidate findings were identified through automated correlation only without individual technical validation.

27. Conclusion

This project successfully demonstrated an end-to-end authorized VAPT workflow within an isolated laboratory environment. PortIntel v2.0 served as the custom reconnaissance and vulnerability-intelligence component of the assessment, performing authorized host discovery, port and service enumeration, and service fingerprinting against the Metasploitable 2 target.

The assessment discovered a broad attack surface consisting of 12 open TCP services, and independent Nmap validation confirmed at least one Critical, exploitable vulnerability: the vsFTPD 2.3.4 backdoor associated with CVE-2011-2523, which was demonstrated to permit root-level command execution within the authorized laboratory.

The project used evidence-based validation throughout rather than accepting scanner output at face value, as illustrated by the treatment of OpenSSH version-correlated Vulners output as candidate intelligence, and by the manual SQL-injection authentication test against Mutillidae, which returned a negative result and was documented as such rather than reported as a confirmed vulnerability.

28. Appendix — Clean Command Reference

```
# 1. Target & Network Verification
ifconfig
ip addr
ping -c 4 192.168.189.130

# 2. PortIntel Installation
git clone https://github.com/kharbashpriyanshu/Port_Intel.git
cd Port_Intel && python3 -m venv .venv && source .venv/bin/activate
python3 -m pip install -e .

# 3. PortIntel Verification & Reconnaissance
portintel version
portintel discover --help
portintel discover --network 192.168.189.0/24
portintel scan --help
portintel scan --target 192.168.189.130

# 4. Independent Nmap Verification & Vulnerability Assessment
nmap -sV -p 21,22,23,25,53,80,111,139,445,512,513,514 192.168.189.130
nmap -sV --script vuln -p 21,22,23,25,53,80,111,139,445 192.168.189.130

# 5. Controlled Validation
# CVE-2011-2523 vsFTPD 2.3.4 backdoor NSE check -> uid=0(root) gid=0(root)
# Manual Mutillidae SQLi test: Username: ' OR '1'='1' -- | Password: test (No bypass)
```